

Tailings

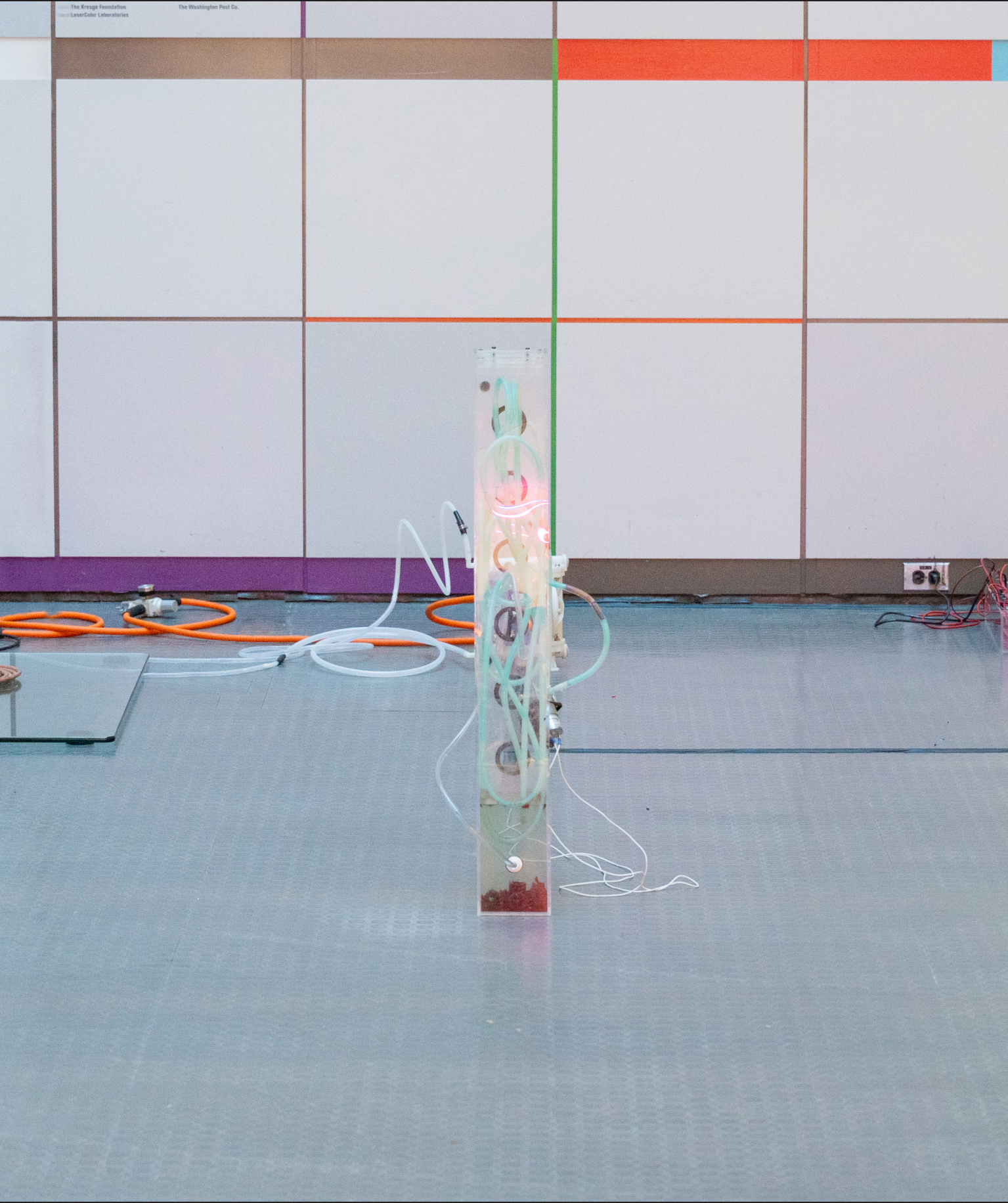
Accumulated mine tailings from copper, cobalt, and coal extraction are increasingly recognized by US industrial policy as latent deposits of rare earth metals—resources now critical to data centers and military technology. The acidic runoff from these mines dissolve and suspend metals in high concentrations, turning industrial waste into a new frontier of extraction. Chemical leaching processes, bioaccumulation systems, and nanoscale filters are being developed to selectively harvest these residues, effectively mining the mine’s own waste. As we begin to consume this second-order material, the model of extraction folds inward, further entering its own synthetic space.

The project grounds itself in the material processes and metabolic flows of Butte, Montana. Once the epicenter of global copper production, Butte is now home to the Berkeley Pit: an acidic lake of industrial waste and popular example of environmental catastrophe at the hands of industrial capitalism. In recent years, U.S. efforts to secure new sources of critical minerals and maintain geopolitical power have reframed Butte’s acidic industrial waste as an unexpected asset. This reorientation of ‘waste’ accumulating in the continental interior as ‘resource’ to be consumed again, is an attempt to bridge a metabolic rift accumulated by the logics of industrial capitalism. As digital infrastructure accelerates, nodes of its supporting material infrastructure surface with unusual clarity in Butte: century-old mine tailings are re-mined for rare earth elements, dissolved copper precipitates onto scrap iron, and a 400-MW data center is built along Silver Bow Creek.

Through staged circulatory models and nanoscale imaging of polluted wastewater, the project explores the logics of industrial capitalism, the mechanical gaze turning inward, loops, false loops, and the components parts

EMAIL
eric_rannestad@gsd.harvard.edu
ericrannestad@gmail.com

BIOGRAPHY
Eric Rannestad is an artist working across sculpture, painting, and digital media. His practice examines models, systems, and technology that humans use to design and control their environment. He received a BA in Studio Art and a BA in Economics from Whitman College and a Master of Design Studies (MDes) at the Harvard Graduate School of Design. He has attended fellowships at the Vermont Studio Center, I-Park Foundation, Sitka Center for Art and Ecology, and more. His commercial projects in cartography and projects in environmental conservation in the American West are a significant influence in his creative practice.



List Visual Arts Center, Massachusetts
Institute of Technology, Installation view of
Tailings, Cambridge, MA, 2025.

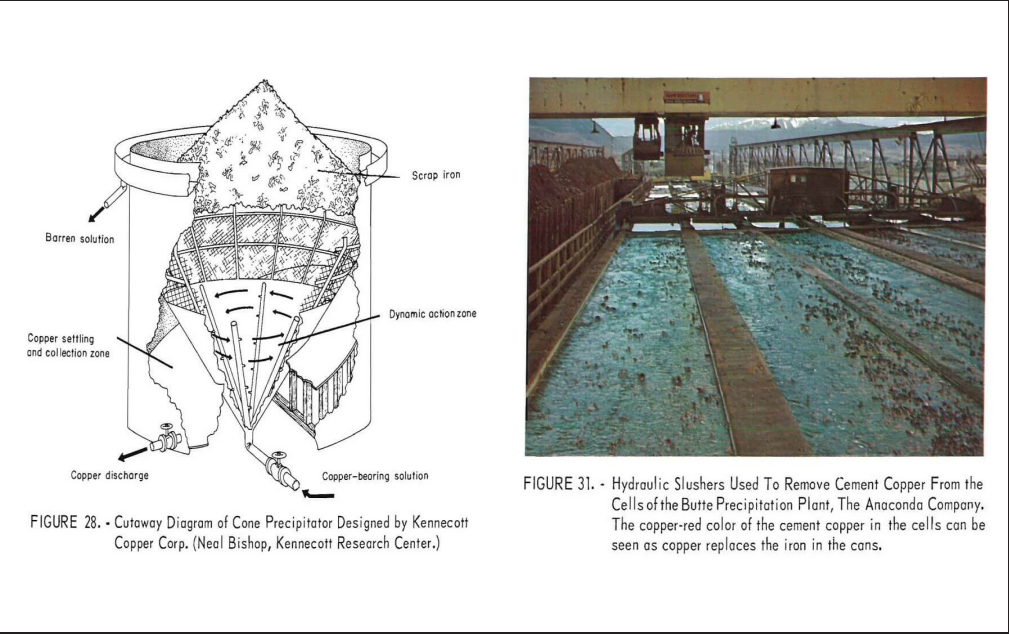


Figure 1. Copper precipitation plant in Butte Montana and example of a historic Precipitator Cone used for the same process in early 20th Century mining operations.

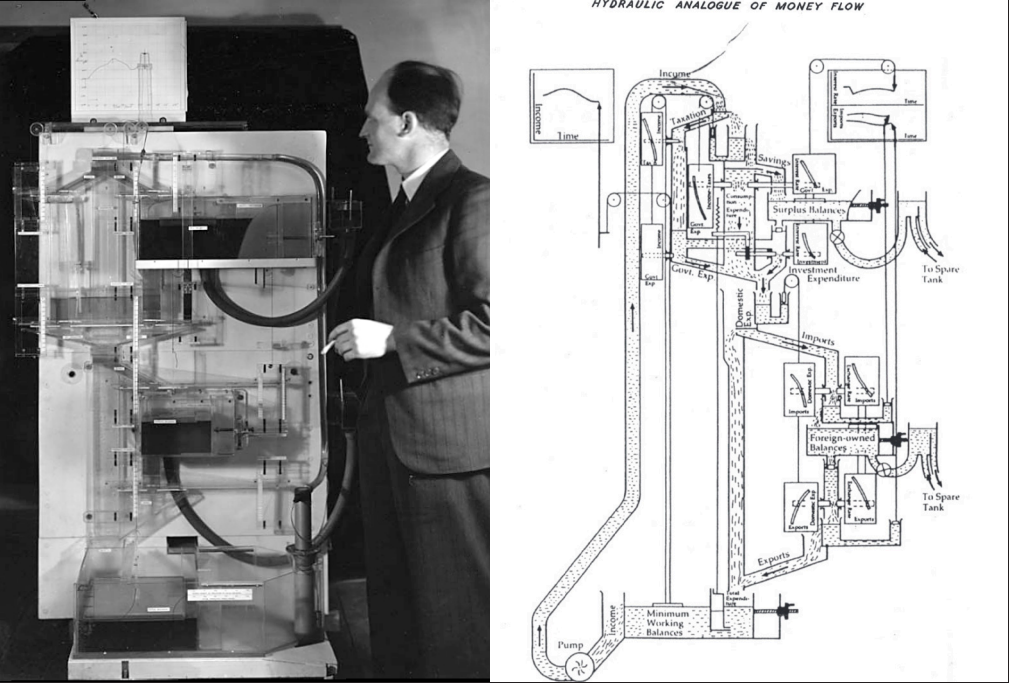
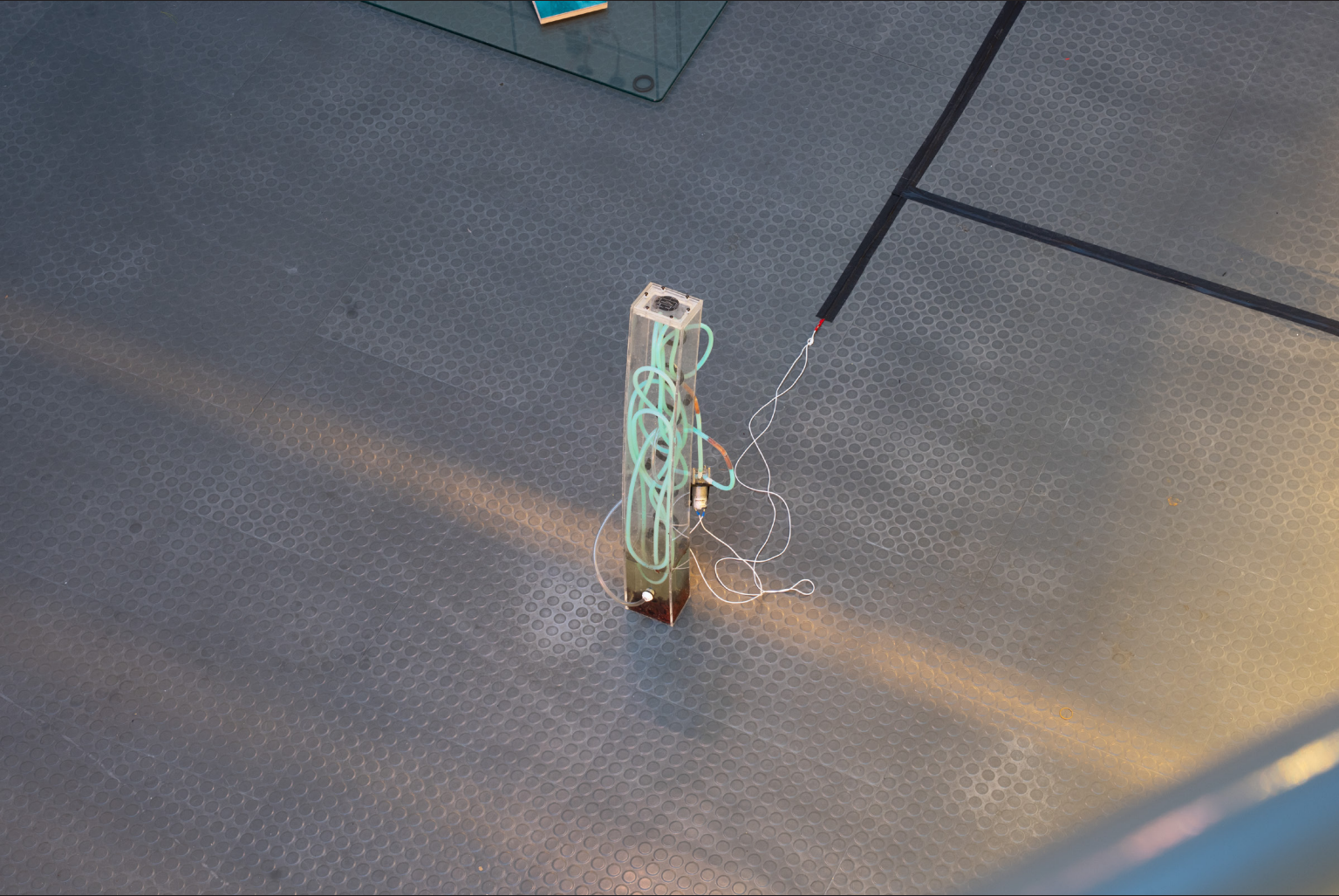
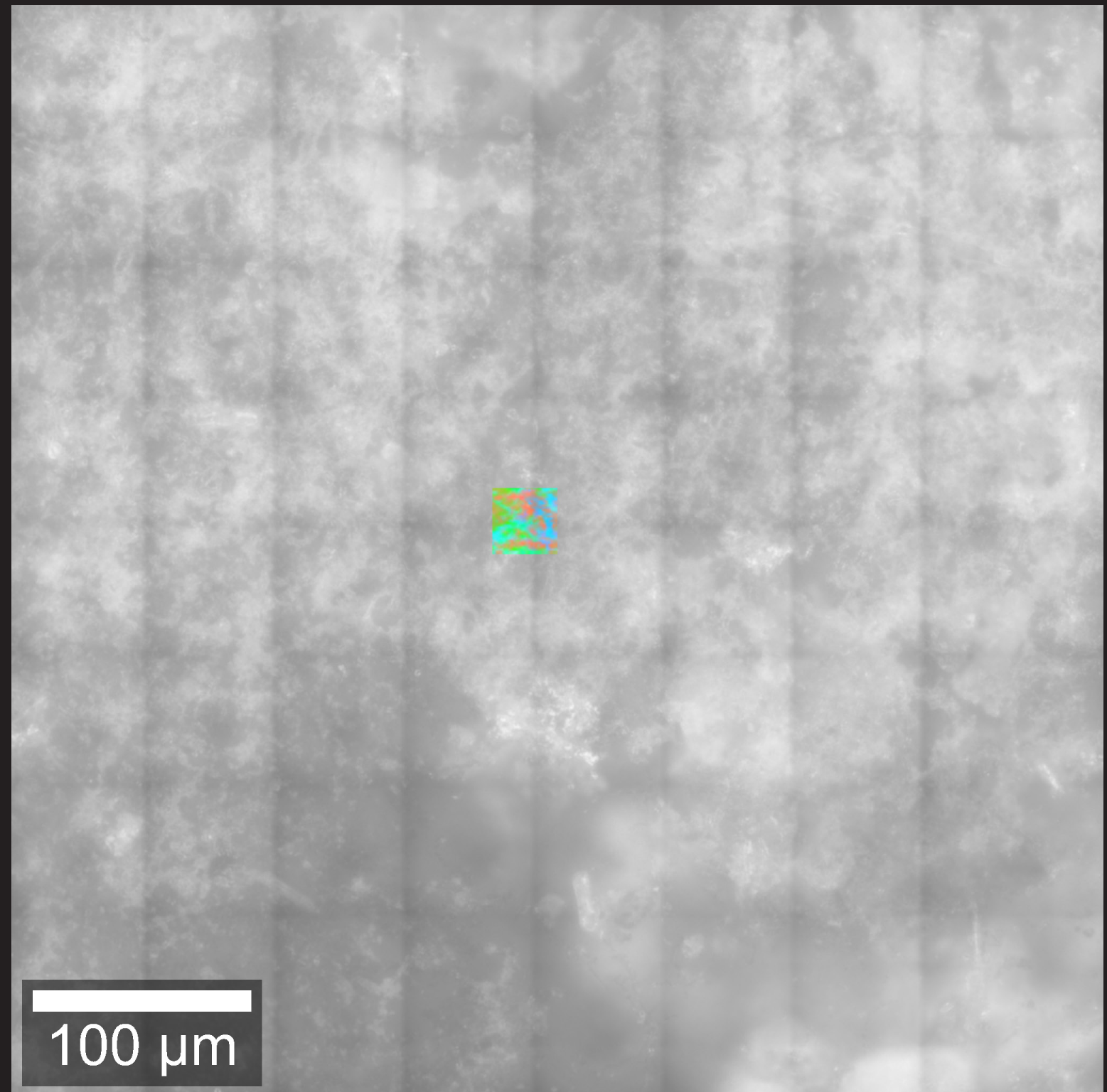
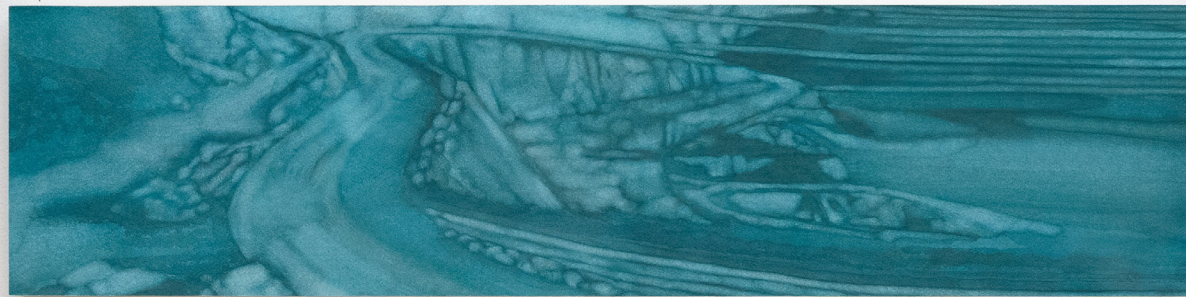


Figure 2. The MONIAC, also known as the Phillips Machine was a device used to model the flow of capital in the economy. It presents a dual image of modernity’s desire to govern the economy as environmental media, and environmental media like the economy. The Accumulator Tanks reference this modeling device and reimagine it within the expanded context of the geographies these economic models hold as given.

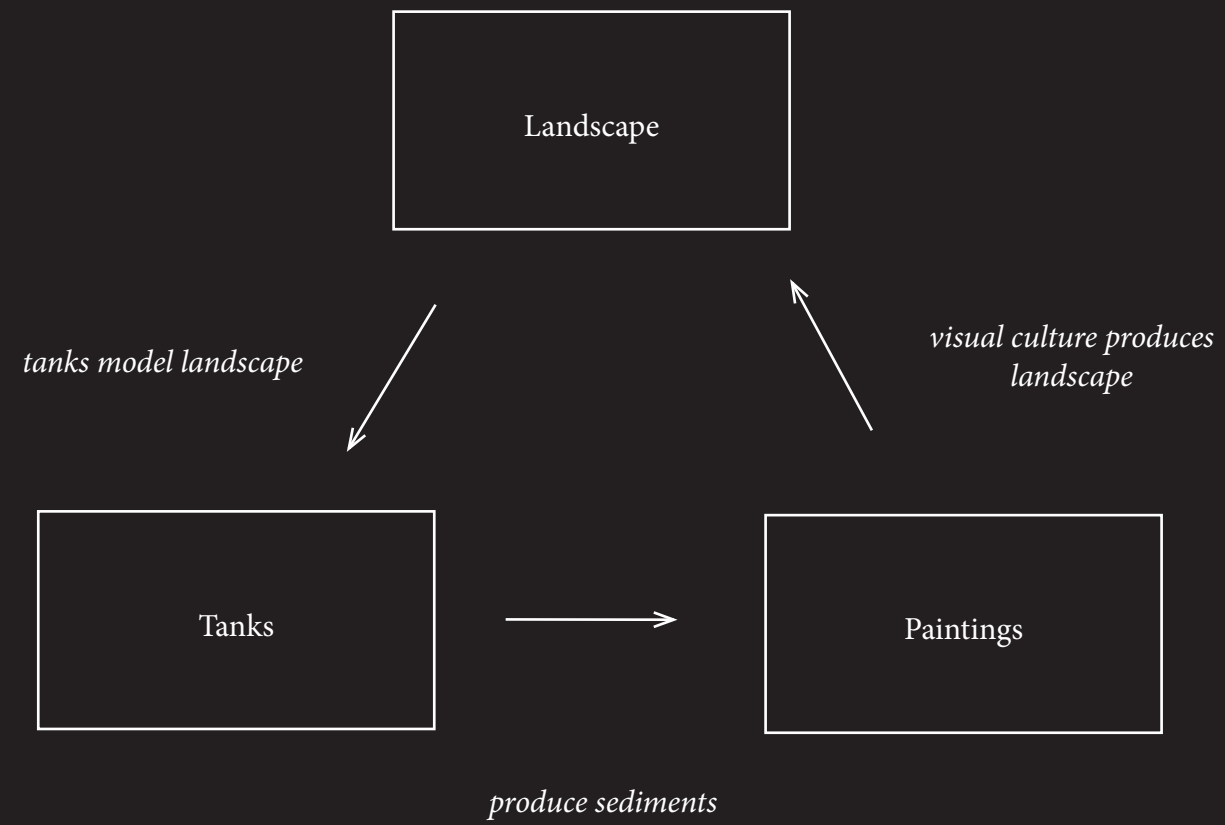


In *Accumulator 01*, an acidic solution circulates through a pair of copper joints. The copper dissolves and becomes a dense blue acid electrolyte. Incrementally, this blue solution is pumped over iron joints on the bottom of the tank. This blue copper solution oxidizes and aggregates as brown metallic clumps of metabolized copper. After running for several days, these sediments are collected as ink and pigment by the artist for use in a series of paints.

Accumulator 01, 2025, copper joints, electrolyte, iron, peristaltic pump, acrylic, silicone, precipitated copper, 40 × 6 × 4 in.



Composite image of chemical composition of industrial waste. Captured by confocal Raman imaging system alpha300 from Witec. Image courtesy of the author.



Since the Tailings Exhibition at MIT List Center for the Visual Arts, this series has grown considerably. These tanks function as conceptual devices for citing the metabolic logics and flows of landscape.

